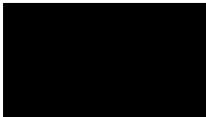


Project N°	Unit	Doc.Type	Material Code	Serial N°	Rev. index
	032	PDS	0910	0028	1

		DIESEL PRODUCT PUMPS		Doc. No.:		Page	3	of	4	
				Client No.:		Rev.:	1			
				Item No.:	P718 (A/B)	Date:	09-Sep-2025			
				Service:	DIESEL PRODUCT PUMPS					
				Client:		Job No.:	-			
Project:										
Location:										
Unit:		Hydrocracking Unit		Area:		032				
ITEM NORMAL OPERATION				P718 (A/B) (9) (20) ⚠						
SPARE ITEM				-						
NUMBER				TWO (1W+1S)						
PUMP TYPE				Centrifugal Pump						
SERVICE				Diesel Product						
PRODUCT HANDLED				Product	Hydrocarbon					
				Auto-ignition temperature	°F	480.0				
				Corrosive/Erosive	(8)	No/No				
				Normal pumping temp. Tp	(16)	°F	330.5			
				Viscosity at Tp	(17)	cP	0.54			
				Vapor pressure at Tp		psia	1.70			
				Specific gravity at Tp			0.758			
				Density at Tp	(15)	lb/ft³	47.29			
				WT% Solids		-	Nil			
				PUMP				Normal flowrate	(11)	GPM
Minimum flowrate	(12)	GPM	464							
Design flowrate		GPM	1021							
Discharge pressure		psig	174.5							
Suction pressure		psig	-2.5							
Differential pressure at design flow		psi	176.9							
Differential head (A)		ft	538.9							
Available NPSH		ft	28.9 (7)							
Normal shaft power (est.)		hp	150.9 (3)							
Normal electrical consumption (est.)		hp	By Vendor (3)							
Design temperature				°F	500 (2)					
Suction design pressure				psig	120.7					
Material of construction (Casing / Impeller)				CS / CS						
Driver type : normal				Electric Motor						
spare				Electric Motor						
Speed control				Yes/No	No					
speed control by :										
Turbine speed control				Manual/Auto	N/A					
Expected speed range				-						
Open valve start-up				Yes/No	Yes (18) ⚠					
GENERAL NOTE										
(A) The choice of a pump having a max. head more than 1.2 time the design head shall be submitted to the Process Dept approval.										

PROCESS DATA SHEET

Project N°	Unit	Doc.Type	Material Code	Serial N°	Rev. index
	032	PDS	0910	0028	1

	DIESEL PRODUCT PUMPS	Doc. No.:		Page	4	of	4
		Client No.:		Rev.:	1		
		Item No.:	P718 (A/B)	Date:	09-Sep-2025		
		Service:	DIESEL PRODUCT PUMPS				
		Client:		Job No.:	-		
		Project:					
		Location:					
Unit:	Hydrocracking Unit	Area:	032				

GENERAL NOTES:

- (B) For utility conditions and site data, please refer to Process Design Basis. For utility condition at vendor battery limit, vendor to consider 7.11 psia additional pressure drop to arrive at vendor battery limit pressure (stringent among min/max/nor to be considered).
- (C) Vendor to provide utility consumption summary with following minimum data per item: Name of utility, consumption rate/quantity, intermittent/continuous, frequency if intermittent, and required operating min/max pressure and temperature.
- (D) Vendor to provide all the auxiliary instrument and equipment datasheet as applicable. Vendor to ensure units of measurement in datasheets as per Process Design Basis.
- (E) For mechanical seal/bearing cooling systems, where cooling water is used, maximum pressure drop allowed across supply and return at pump skid battery limit is 9.95 psia. Vendor shall ensure backflush provision and a bleed point with valve and blind at upsetstream of isolation valve on cooling water return line. Cooling water composition is as specified in the Process Design Basis.

NOTES:

- Maximum allowable sound pressure level for pump package : 85 dB(A) at 3 ft.
- Minimum Design Metal Temperature = 10 °F
- To be confirmed by vendor.
- Vendor to limit the shut off pressure for the pump to 350 psig.
- Seal plan type for pump to be confirmed by Rotary specialist / Vendor.
- API standard 610 (latest edition) applies unless otherwise indicated.
- NPSH available shall be at least 3 ft above NPSH required. NPSHa is specified at pump centerline assuming pump centerline is at 3 ft from grade.
- Service is hazardous and non toxic.
- Pump type: Horizontal. Pump in continuous operation.
- Respiratory Exposure Hazard Class for P718 is R4.
- Normal flow for the pump is for HMB Case 1 - High Conversion SOR(Rev5).
- Minimum flow for the pump is 50% of normal flow.
- Selected pump shutoff pressure including API margin shall not exceed estimated pump shutoff pressure indicated in datasheet.
- Indicated flow (Rated / Normal / Minimum) is process required flow and does not include pump warm-up flow. Pump Warm-up line is provided with Restriction Orifice.
- Vendor to confirm standby pump warmup-flowrate and shall include in specified pump rated capacity to arrive at final rated capacity of pump. Rated head shall be guaranteed without any -ve margin at final rated capacity of pump. Restriction Orifice on Warm-up line shall be loose supply by pump vendor.
- Pump vendor to also advice minimum circulation flowrate for the pump.
- Maximum operating density is 47.4 lb/ft3.
- Pumping temperature of the fluid ranges from 324.92 to 352.47 °F.
- Viscosity of the fluid ranges from 0.47 to 0.56 cP.
- Motor to be designed for end of curve operation considering start-up/off-spec case as specified in Note-20.
- Vendor to provide the pump curve for parallel pump operation.

Case	Total Flow (GPM)	Suction Pressure (psig)	Discharge Pressure (psig)	Differential Pressure (psi)	Differential Head (ft)	Available NPSH (ft)	Shaft Power (hp)	Density (lb/ft ³)	Viscosity (cP)	Temperature (°F)	Operation
Off-Spec. to PGO	798	-2.4	202.8	205.2	625	29.2	68.4	47.3	0.54	330.5	Two (Parallel)